Lecture 13 (*March 17*)

Reading: Section 4.3
Viewing: Lecture 16

A big computational application of projections is **least squares** approximation, which we will now delve into. Mathematically, the problem involves a system of equations for given A and $\boldsymbol{b}$:

$$A\boldsymbol{v} = \boldsymbol{b} \tag{107}$$

which does not have any solutions, because the number of equations is greater than the number of variables (mathematically, think of A as an $m \times n$ matrix with $m > n$). Therefore, the task is to find a value of $\boldsymbol{v}$ which makes $A\boldsymbol{v}$ as close as possible to $\boldsymbol{b}$. In other words, the **error** vector:

$$\boldsymbol{e} = \boldsymbol{b} - A\boldsymbol{v}$$

should be made as "small" as possible. Since we are dealing with vectors, "small" means that the length of the vector should be as small as possible. Therefore, the problem is to find $\boldsymbol{v}$ such that:

$$||\boldsymbol{e}|| = ||\boldsymbol{b} - A\boldsymbol{v}|| \tag{108}$$

is as small as possible, for given A and $\boldsymbol{b}$. But what does this mean geometrically? As $\boldsymbol{v}$ ranges over all possible vectors, $A\boldsymbol{v}$ ranges over the column space $C(A)$. If $\boldsymbol{b}$ were already in $C(A)$, then the equation (107) could be satisfied with no error. But in general, $\boldsymbol{b} \notin C(A)$, in which case:

$$||\boldsymbol{b} - A\boldsymbol{v}|| \geq \Big(\text{distance from } \boldsymbol{b} \text{ to the subspace } C(A)\Big) = ||\boldsymbol{b} - \boldsymbol{p}||$$

where $\boldsymbol{p}$ is the orthogonal projection of $\boldsymbol{b}$ onto the subspace $C(A)$. Therefore, finding $\boldsymbol{v}$ which minimizes the quantity (108) actually involves two steps:

- find the orthogonal projection $\boldsymbol{p}$ of $\boldsymbol{b}$ onto the subspace $C(A)$
- solve the equation $\boxed{A\boldsymbol{v} = \boldsymbol{p}}$ (which can be done, since $\boldsymbol{p} \in C(A)$)

We already know the answer to the first question from (102):

$$\boldsymbol{p} = A(A^T A)^{-1} A^T \boldsymbol{b}$$

The answer to the second question is also manifest, because a solution to the boxed equation is:

$$\boldsymbol{v} = (A^T A)^{-1} A^T \boldsymbol{b} \tag{109}$$

Recall the fact (104): in order for the matrix $A^T A$ to be invertible and for the formula above to make sense, we need the matrix A to have independent columns. This can always be arranged by removing some dependent columns from the matrix A: for example, if the last column of A is a linear combination of the other ones, then we can simply remove the last column of A and the last entry of $\boldsymbol{v}$, and the task of approximating equation (107) as well as possible remains the same.

Remark. *You could also have arrived at formula* (109) *from calculus. Specifically, it's obvious that* $||\boldsymbol{b} - A\boldsymbol{v}||$ *is minimum when* $||\boldsymbol{b} - A\boldsymbol{v}||^2$ *is minimum. The usual formulas give us:*

$$||\boldsymbol{b} - A\boldsymbol{v}||^2 = (\boldsymbol{b} - A\boldsymbol{v})^T(\boldsymbol{b} - A\boldsymbol{v}) = \boldsymbol{b}^T\boldsymbol{b} - 2(A\boldsymbol{v})^T\boldsymbol{b} + (A\boldsymbol{v})^T(A\boldsymbol{v}) = \boldsymbol{b}^T\boldsymbol{b} - 2\boldsymbol{v}^TA^T\boldsymbol{b} + \boldsymbol{v}^TA^TA\boldsymbol{v}$$

Now, the latter quantity is minimum when its derivative with respect to $\boldsymbol{v}$ *vanishes (there is a mathematical notion of taking derivative with respect to a vector of variables, and it behaves just like derivative with respect to a single variable), which explicitly states that:*

$$0 = \frac{\partial(\boldsymbol{b}^T\boldsymbol{b} - 2\boldsymbol{v}^TA^T\boldsymbol{b} + \boldsymbol{v}^TA^TA\boldsymbol{v})}{\partial \boldsymbol{v}} = -2A^T\boldsymbol{b} + 2A^TA\boldsymbol{v}$$

Solving the equation above for $\boldsymbol{v}$ *gives precisely* (109).

Let's do a concrete example. Find numbers x, y, z such that the vector:

$$x\begin{bmatrix}1\\1\\0\end{bmatrix} + y\begin{bmatrix}0\\1\\1\end{bmatrix} + z\begin{bmatrix}1\\2\\1\end{bmatrix} \quad \text{is as close to} \quad \begin{bmatrix}1\\2\\3\end{bmatrix} \quad \text{as possible} \tag{110}$$

A straightforward application of the analysis above would have us set $A = \begin{bmatrix}1 & 0 & 1\\1 & 1 & 2\\0 & 1 & 1\end{bmatrix}$ and apply formula (109). But the columns of A are not independent: in fact, the third column is the sum of the first two. This is reflected in the fact that equation (110) reads:

$$x\begin{bmatrix}1\\1\\0\end{bmatrix} + y\begin{bmatrix}0\\1\\1\end{bmatrix} + z\left(\begin{bmatrix}1\\1\\0\end{bmatrix} + \begin{bmatrix}0\\1\\1\end{bmatrix}\right) = (x+z)\begin{bmatrix}1\\1\\0\end{bmatrix} + (y+z)\begin{bmatrix}0\\1\\1\end{bmatrix}$$

Since x, y, z can be chosen arbitrarily, the problem us equivalent to finding a and b such that:

$$\text{the vector} \quad a\begin{bmatrix}1\\1\\0\end{bmatrix} + b\begin{bmatrix}0\\1\\1\end{bmatrix} \quad \text{is as close to} \quad \begin{bmatrix}1\\2\\3\end{bmatrix} \quad \text{as possible}$$

In matrix form, this is translated as:

$$\underbrace{\begin{bmatrix}1 & 0\\1 & 1\\0 & 1\end{bmatrix}}_{A}\underbrace{\begin{bmatrix}a\\b\end{bmatrix}}_{\boldsymbol{v}} - \underbrace{\begin{bmatrix}1\\2\\3\end{bmatrix}}_{\boldsymbol{b}} \text{ is as small as possible} \tag{111}$$

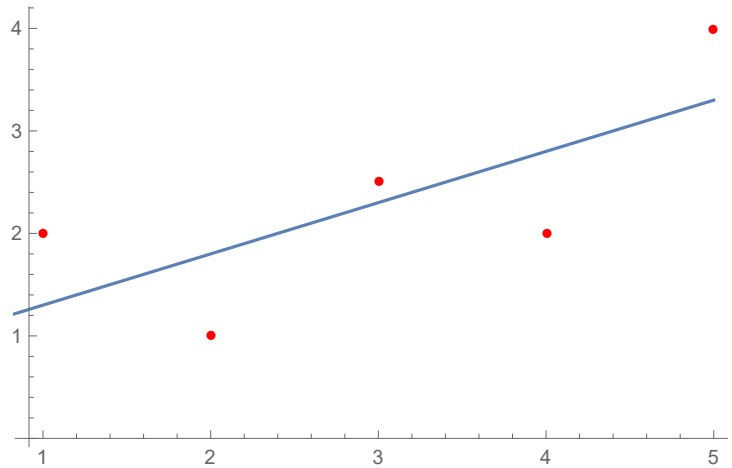
Now the columns are independent, so we can apply formula (109). We have:

$$A^TA = \begin{bmatrix}1 & 1 & 0\\0 & 1 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\1 & 1\\0 & 1\end{bmatrix} = \begin{bmatrix}2 & 1\\1 & 2\end{bmatrix} \quad \Rightarrow \quad (A^TA)^{-1} = \frac{1}{3}\begin{bmatrix}2 & -1\\-1 & 2\end{bmatrix}$$

and so (109) implies that the difference (111) is minimized for:

$$\begin{bmatrix}a\\b\end{bmatrix} = \frac{1}{3}\begin{bmatrix}2 & -1\\-1 & 2\end{bmatrix}\begin{bmatrix}1 & 1 & 0\\0 & 1 & 1\end{bmatrix}\begin{bmatrix}1\\2\\3\end{bmatrix} = \frac{1}{3}\begin{bmatrix}2 & 1 & -1\\-1 & 1 & 2\end{bmatrix}\begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}\frac{1}{3}\\\frac{7}{3}\end{bmatrix}$$

We now turn to what is perhaps the standard application of least squares approximation, which will explain its name. Suppose you have a graph which seeks to represent a time-dependent quantity $y(t)$. You measure this quantity at times $t_1, \ldots, t_m$ and get values $y_1, \ldots, y_m$. Mark the data points $(t_1, y_1), \ldots, (t_m, y_m)$ on a graph, as follows (say $m = 5$ data points in the picture below):



The goal is to "fit" a straight line L between these data points as closely as possible. Mathematically, the word "closely" means that the following condition must be satisfied:

the sum of squares of the vertical distances from the data points to L is minimum

which explains the terminology "least squares". So how do we do find this best fit line? To choose a line is the same thing as to choose two numbers a and b, so that the equation of the line is $y(t) = a + bt$. Then the sum of squares that we need to minimize is the quantity:

$$(y_1 - a - bt_1)^2 + \cdots + (y_m - a - bt_m)^2 \tag{112}$$

where $t_1, \ldots, t_m$ and $y_1, \ldots, y_m$ are given to you by your experiment. How to convert this into a linear algebra problem? If the line were to pass precisely through our data points, this would require the following system of equations to be satisfied (where a and b are the unknowns):

$$\begin{cases} a + bt_1 = y_1 \\ \ldots \\ a + bt_m = y_m \end{cases} \qquad \text{or, equivalently} \qquad \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}$$

In practice, we will not be able to solve this system on the nose, but the condition that the quantity (112) is minimized is exactly the condition that the square of (108) is minimized, for:

$$A = \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix} \quad \text{and} \quad \boldsymbol{b} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}$$

We have:

$$A^T A = \begin{bmatrix} 1 & \ldots & 1 \\ t_1 & \ldots & t_m \end{bmatrix} \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix} = \begin{bmatrix} m & t_1 + \cdots + t_m \\ t_1 + \cdots + t_m & t_1^2 + \cdots + t_m^2 \end{bmatrix} \quad \Rightarrow$$

$$\Rightarrow \quad (A^TA)^{-1} = \frac{1}{m(t_1^2+\cdots+t_m^2)-(t_1+\cdots+t_m)^2}\begin{bmatrix} t_1^2+\cdots+t_m^2 & -t_1-\cdots-t_m \\ -t_1-\cdots-t_m & m \end{bmatrix}$$

So formula (109) tells us that the choice of a and b which gives the best straight line fit is:

$$\begin{bmatrix} a \\ b \end{bmatrix} = \frac{1}{m(t_1^2+\cdots+t_m^2)-(t_1+\cdots+t_m)^2}\begin{bmatrix} t_1^2+\cdots+t_m^2 & -t_1-\cdots-t_m \\ -t_1-\cdots-t_m & m \end{bmatrix}\begin{bmatrix} 1 & \dots & 1 \\ t_1 & \dots & t_m \end{bmatrix}\begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}$$

and therefore:

$$\begin{bmatrix} a \\ b \end{bmatrix} = \frac{1}{m(t_1^2+\cdots+t_m^2)-(t_1+\cdots+t_m)^2}\begin{bmatrix} \sum_{i=1}^n t_i(t_i-t_1) & \dots & \sum_{i=1}^n t_i(t_i-t_m) \\ \sum_{i=1}^n (t_1-t_i) & \dots & \sum_{i=1}^n (t_m-t_i) \end{bmatrix}\begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}$$

The fact that the fraction in the formula above is non-zero is a consequence of the Cauchy-Schwarz inequality, which states that $m(t_1^2+\cdots+t_m^2) > (t_1+\cdots+t_m)^2$ if $t_1, \dots, t_m$ are all distinct numbers.